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Deepfakes:
Can You Still Trust What You See on the Internet?

TECHNOLOGY | PSYCHOLOGY | CYBERSECURITY

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Founder, ThinkHub

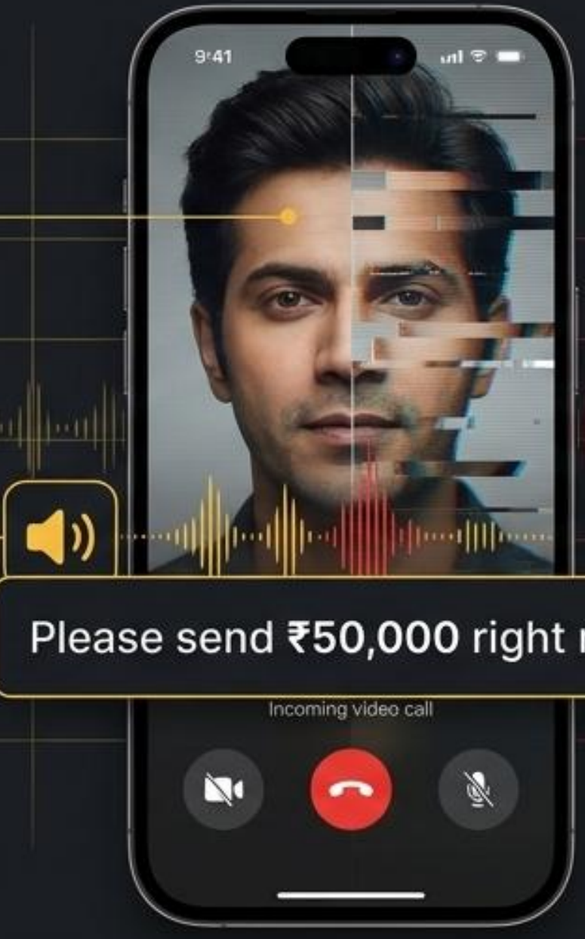


The Video Call You Cannot Trust

What if they never made this call?

A familiar face.

A familiar voice.



An urgent situation.

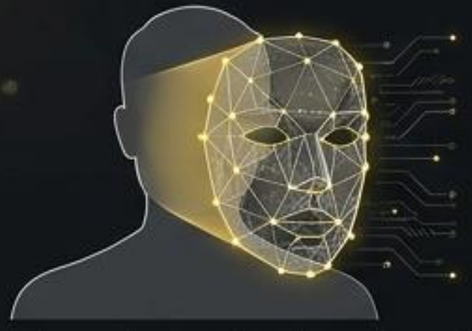
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WHAT EXACTLY IS A DEEPFAKE?

Synthetic or manipulated media produced using AI to **alter identity or fabricate events**.

FACE REPLACEMENT



Digital mask overlays and replaces target identity.

CLONED SPEECH



AI generates realistic voice from audio samples.

ALTERED LIPS



Visuals synced to a distinct, fabricated audio track.

GENERATED SCENE



Entire environments built from AI-driven prompts.

Synthetic media is not automatically malicious. Harm depends on context, consent, and the claim being made.

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FOUR WAYS AI CAN FAKE REALITY

Synthetic or manipulated media produced using AI to alter identity or fabricate events.

1. FACE SWAP



Digital mask overlays and replaces target identity.

2. VOICE CLONE



AI generates realistic voice from audio samples.

3. FACIAL REENACTMENT



Visuals synced to a distinct, fabricated audio track.

4. VIDEO GENERATION



Entire environments built from AI-driven prompts.

These methods can be combined into one deceptive hybrid.

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INSIDE THE DEEPFAKE PIPELINE

1. INPUT



Image, Video, Audio, or Text Prompts

2. AI MODEL



Learns patterns / Generates identity

3. SYNTHESIS



Coordinates timing, expression, and motion

4. POST-PROCESSING



Editing, rendering, formatting

5. DISTRIBUTION



Social media, fabricated evidence, urgent calls

TEMPORAL CONSISTENCY
Ensuring frames form a coherent, continuous sequence

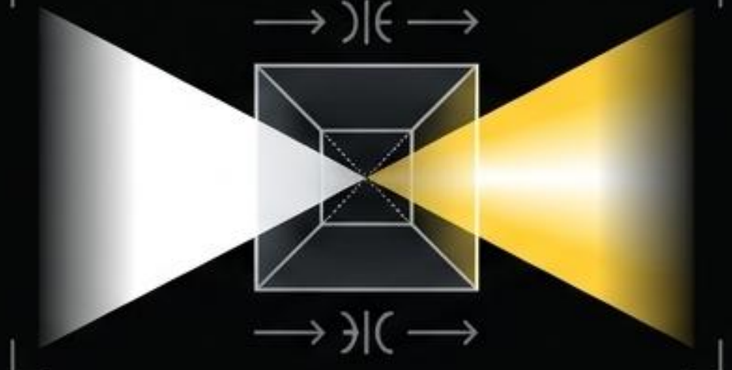
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THE AI ENGINES BEHIND THE ILLUSION



GENERATIVE ADVERSARIAL NETWORKS (GANs)

An adversarial loop to perfect synthetic outputs.



AUTOENCODERS

Encoding input into a compressed representation, then reconstructing it.



DIFFUSION MODELS

Learning to generate data by reversing a noise-addition process.

Modern implementations often combine these models with transformers and neural speech generation.

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WHY OUR BRAINS BELIEVE THE FAKE

FAMILIARITY

A face we recognize feels instantly trustworthy.

EMOTION

Fear or excitement bypasses logical scrutiny.



URGENCY

Less time to pause and verify.

SOCIAL PROOF

Viral metrics (likes/shares) are mistaken for truth.

CONFIRMATION BIAS

We accept what supports our existing beliefs.

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WHEN A FAKE BECOMES A REAL CRIME



1. GATHER

Identify target routines via public information.

2. IMPERSONATE

Clone voice or face of a trusted contact.

3. CREATE URGENCY

Fabricate an emergency or tight deadline.

4. REQUEST

Demand money, account access, or credentials.

5. EXPLOIT DELAY

Move assets before independent verification occurs.

FBI WARNING

May 2025: FBI documents campaigns using AI-generated voices to impersonate senior officials. The attack only needs to fool one specific recipient.

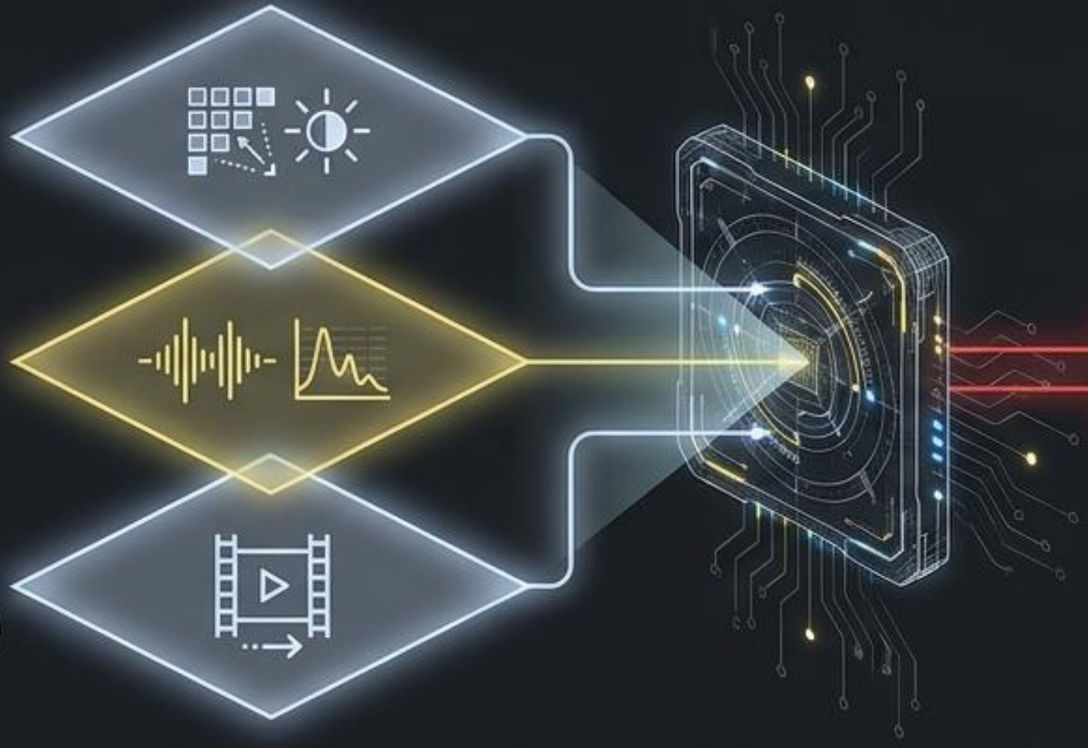
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CAN AI CATCH AI?

LAYER 1: VISUAL
(PIXELS & LIGHTING)

LAYER 2: AUDIO
(SPECTRAL PATTERNS)

LAYER 3: TEMPORAL
(FRAME-TO-FRAME MOTION)



FALSE NEGATIVE:
UNSEEN GENERATORS OR EXTREME COMPRESSION EVADE DETECTION.

FALSE POSITIVE:
AUTHENTIC MEDIA INCORRECTLY FLAGGED AS MANIPULATED.

A DETECTOR SCORE IS AN INITIAL SIGNAL, NOT DEFINITIVE PROOF OF REALITY.

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THE EVIDENCE PROBLEM: REAL, FAKE, OR MISLEADING?

Y-Axis:
THE MEDIA
(Authentic Recording vs. Synthetic/Manipulated)

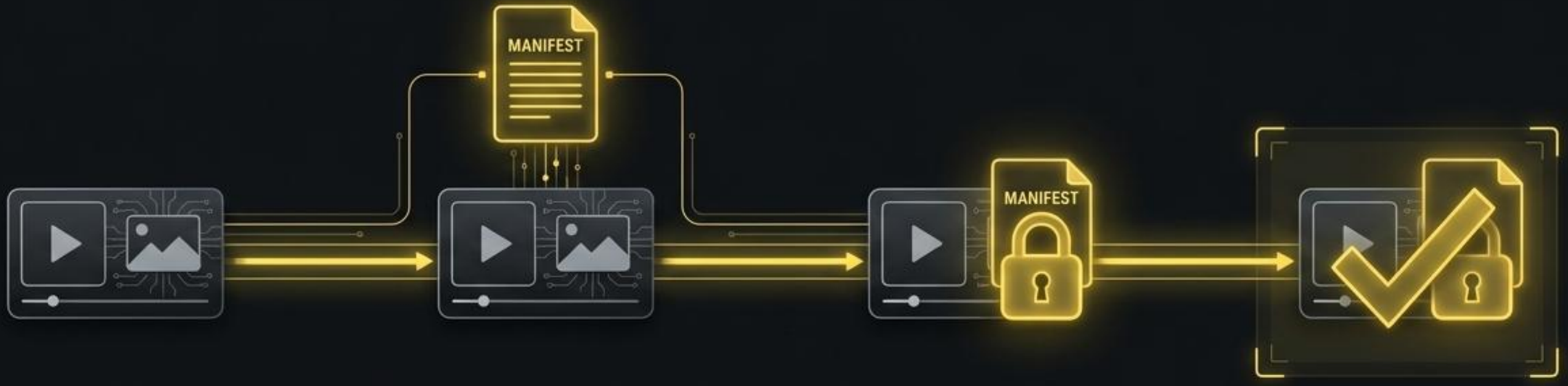


X-Axis:
THE CLAIM (Accurate Context vs. Misleading/Deceptive)

THE LIAR'S DIVIDEND: Convincing fakes make it easier for people to dismiss genuine, damaging evidence.

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CONTENT CREDENTIALS: A DIGITAL PAPER TRAIL



1. CREATION

Media captured by camera or supported AI application.

2. MANIFEST

Provenance assertions (origin/edits) are recorded.

3. BINDING

A cryptographic hash and digital signature fuse the manifest to the asset.

4. VERIFICATION

A compatible verifier checks the lock's integrity.

A VALID CREDENTIAL PROVES ORIGIN AND HISTORY—IT DOES NOT INDEPENDENTLY PROVE THE DEPICTED EVENT IS TRUTHFUL.

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WATERMARKS, SIGNATURES, AND THEIR LIMITS

DIGITAL WATERMARK

What it does: Embeds a detectable signal into the media.

Limitation: Robustness varies; compression can destroy the signal.



DIGITAL SIGNATURE

What it does: Uses cryptographic keys to help detect unauthorized

Limitation: Only as trustworthy as the key and signer identity.



PROVENANCE (C2PA)

What it does: Records cryptographically verifiable history of origin and changes.

Limitation: Coverage is incomplete; missing credentials do not prove a fake.



INTEGRITY IS NOT THE SAME AS TRUTH.

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INDIA'S RESPONSE TO SYNTHETIC MEDIA

NOV 27, 2024

CERT-In publishes 'Deepfakes: Threats and Countermeasures' advisory.

Addresses defined categories of realistic AI-generated/altered audio-visual content.

FEB 10, 2026

Government issues G.S.R. 120(E), amending IT Rules (2021).

Introduces provisions for labelling and provenance mechanisms.

FEB 20, 2026

Amendments regarding synthetically generated information take effect.

Crucial distinction: Not all AI content is prohibited; depends on unlawful use, scope, and specific exceptions.

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THE THINK FRAMEWORK: VERIFY BEFORE YOU TRUST

T — **Take a pause**
Resist panic, outrage, or artificial urgency.

H — **Hunt for the source**
Find the original upload, avoid forwarded screenshots.



I — **Independently verify**
Call the person using a saved, trusted number.

K — **Keep private**
Do not forward, fund, or approve until verified.

N — **Notice context**
Check dates, provenance credentials, and reporting.

India Cybercrime Portal (cybercrime.gov.in) | Helpline: 1930

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THE FUTURE OF DIGITAL IDENTITY

[High-consequence decisions require multiple independent signals, not just visual recognition.]



LAYER 4 (RISK CHECKS)
Escalation protocols and independent confirmation channels.

LAYER 3 (AUTHORIZATION)
Are they permitted to do this?
(Dual approvals, established procedures).

LAYER 2 (AUTHENTICATION)
Trusted credentials, cryptographic signatures, registered devices.

BASE LAYER (IDENTITY CLAIM)
Who is requesting this?
(The face/voice on screen).

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“In the age of deepfakes, the question isn't just whether you can trust what you see. It's whether you know how to verify what you believe.”



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Before you share, verify.

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THINK. LEARN. GROW.

THANK YOU!

The future belongs to those who build it continuously.

START TODAY.
Take the first step.

STAY CONSISTENT.
Small steps, big results.

BUILD ONE PROJECT
at a time.



PRESENTED BY

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FOUNDER, THINKHUB

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